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A VARIATIONAL PRINCIPLE FOR THE LORENTZ CONDITION AND RESTRICTION OF THE DOMAIN OF PATH INTEGRATION IN NON-ABELIAN GAUGE THEORY

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A variational principle generating an additional Lorentz condition in gauge field theory is considered. It is shown that an absolute minimum of the functional varied is achieved on an appropriate set. Properties of the collection of fields corresponding to such minima are described. A path integral over this set of fields is considered.

1. Passage beyond the framework of perturbation theory in quantization of gauge fields leads to new and interesting mathematical questions. Thus, in the work of Gribov [1] it is shown that in using the Lorentz gauge for quantization of the Yang–Mills field by means of a path integral the domain of integration over the potentials is to be restricted in an appropriate manner. This is necessary in order that each orbit of the gauge group be considered only once. As noted in [1], the change of the domain of integration proposed there suppresses the low-frequency part of the field.

The purpose of the present work is to obtain additional information on the correct domain

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of path integration in the Lorentz gauge. This can be done by applying a variational principle from which the Lorentz condition in four-dimensional Euclidean space follows [2]. There is an analogous variational principle in the case of the Coulomb gauge condition in three-dimensional space. A number of consequences of this type of variational principle are considered below.

2. We consider a gauge field $A_\mu = A^a_\mu(x)T^a = -A^{a+}_\mu$ with structure group $\mathcal{G}$ = SU(N) in D-dimensional Euclidean space. All arguments carry over easily to the case of any gauge group. In the basic representation we set $\mathrm{tr}(T^aT^b) = -(\frac{1}{2})\delta^{ab}$. For arbitrary matrix vectors $\mathcal{B}_\mu$, $\mathcal{B}'_\mu$ we write $(\mathcal{B},\mathcal{B}') = (\mathcal{B}_\mu,\mathcal{B}'_\mu) = 2\,\mathrm{tr}(\mathcal{B}^+_\mu \mathcal{B}_\mu)$. In particular, $(A,A) = A^a_\mu A^a_\mu = -2\,\mathrm{tr}(A_\mu A_\mu)$. We denote by $D_\mu(A)$ the covariant derivative in the field A_μ.

The action

$$S = \frac{1}{4g^2}\int d^Dx\, \mathcal{G}^a_{\mu\nu}\, \mathcal{G}^a_{\mu\nu},$$

where $\mathcal{G}_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$ is invariant under gauge transformations

$$A_\mu \to A^u_\mu = u^+ A_\mu u + u^+\partial_\mu u, \tag{1}$$

where $u^+u = 1$, $\det u = 1$. The coupling constant g is introduced so that the potentials have the dimension of momentum for any D. We restrict attention to fields belonging to the space L_2, i.e., satisfying the condition

$$|A|^2 = \int d^Dx\,(A,A) < \infty .$$

We consider the functional

$$f_A[u] = |A^u|^2 \tag{2}$$

of the matrix function u(x) for a fixed field A_μ.

Proposition 1. If u_0 is an extremal of the functional (2), then the field $A^{u_0}_\mu$ is transverse; the second variation of the functional $f_A[u]$ is a quadratic form in the operator $M = -\partial_\mu D_\mu(A^{u_0})$.

Proof. We set $u = u_0 \exp(-\omega)$, where ω is an infinitely small anti-Hermitian matrix, and compute the linear and quadratic terms of the expansion of the functional $f_A[u_0 \exp(-\omega)]$ in a series in ω:

$$\delta f_A = -2\int d^Dx\,(\partial_\mu\omega, A^{u_0}_\mu), \tag{3}$$

$$\delta^2 f_A = \int d^Dx\,(\partial_\mu\omega, D_\mu(A^{u_0})\,\omega) . \tag{4}$$

If the function $A^{u_0}_\mu$ is differentiable, then the equality $\delta f_A = 0$ leads to the Lorentz condition

$$\partial_\mu A^{u_0}_\mu = 0. \tag{5}$$

In this case the expression (4) takes the form

$$\delta^2 f_A = \int d^D x\,(\omega, M\omega)\,.$$

If the function $A_\mu^{u_0} \in L_2$ is not differentiable, then Proposition 1 is to be understood in the weak sense of (3), (4).

We note that the operator M is Hermitian if equality (5) is satisfied. We point out also that the condition of the background-gauge

$$\mathcal{D}_\mu(\mathcal{B})(A_\mu^{u_0} - \mathcal{B}_\mu) = 0$$

can be obtained by varying with respect to u the integral

$$\int d^D x\,(A^u - \mathcal{B}, A^u - \mathcal{B})\,.$$

We henceforth assume that $D \geq 3$. We shall determine a sufficiently broad class of functions u(x) not taking potentials out of the space L_2 under the transformations (1). Let $A_\mu \in L_2$, $A_\mu^u \in L_2$, $\ell = u\,\partial_\mu u^+$. Then

$$|A^u|^2 = |A - \ell|^2$$

and

$$|u|_1^2 = \int d^D x(\partial_\mu u, \partial_\mu u) = 2\int d^D x\,\mathrm{tr}(\partial_\mu u^+\,\partial_\mu u) = |\ell|^2\,, \tag{6}$$

whereby, because of the triangle inequality,

$$|u|_1 = |\ell| \leq |A| + |\ell - A| = |A| + |A^u| < \infty\,. \tag{7}$$

This relation motivates the following construction.

Let C_0^∞ be the space of a compactly supported, complex matrix-valued $N \times N$ functions on R^D having continuous derivatives of all orders. Each function $w \in C_0^\infty$ vanishes outside a bounded region. For $D \geq 3$ functions $w \in C_0^\infty$ satisfy the inequality

$$\int \frac{d^D x}{x^2}(w, w) \leq \frac{4}{(D-2)^2}\,|w|_1^2\,, \tag{8}$$

where

$$|w|_1^2 = \int d^D x(\partial_\mu w, \partial_\mu w)\,. \tag{9}$$

The completion L_2^1 of the space C_0^∞ in the norm (9) consists of all functions w for which

$$\|w\|^2 = \int \frac{d^D x}{x^2}(w, w) + \int d^D x\,(\partial_\mu w, \partial_\mu w)\,. \tag{10}$$

L_2^1 is a Hilbert space equipped with the norm (10). We shall call the Sobolev group $\mathfrak{K}(\mathcal{G})$ of gauge transformations the collection of all matrix-valued functions u(x) for which $(u - I) \equiv w \in L_2^1$ and $u^+u = I$, $\det u = 1$ almost everywhere with respect to Lebesgue measure on R^D. If $u \in \mathfrak{K}(\mathcal{G})$ then the derivatives $\partial_\mu u = \partial_\mu w$ exist almost everywhere, and the function $w = u - I$ is subject to conditions (8) and (10).

Transformations belonging to the Sobolev group do not take potentials out of L_2. Indeed, suppose $A_\mu \in L_2$, $u \in K(\mathcal{G})$, $w \equiv u - I$. Then by equality (10) $\partial_\mu u = \partial_\mu w \in L_2$. Further, $uu^+ = I$ almost everywhere, and hence

$$|Au|^2 = \int d^D x\,(A, Auu^+) = \int d^D x (A, A) < \infty\ ,$$

i.e., $(A_\mu u) \in L_2$. Hence $(\partial_\mu u + A_\mu u) \in L_2$ and

$$|A^u|^2 = \int d^D x (\partial_\mu u + A_\mu u, uu^+ (\partial_\mu u + A_\mu u)) = |\partial_\mu u + A_\mu u|^2 < \infty,$$

so that $A_\mu^u \in L_2$.

The Sobolev group forms the class of transformations we need. This group does not contain global transformations by matrices not depending on the coordinates, since for such matrices $\|u - I\| = \infty$ if $u \neq I$.

Proposition 2. If $A_\mu \in L_2$, then the functional $f_A[u] \equiv |A^u|^2$ has an absolute minimum on the Sobolev group $K(\mathcal{G})$.

Proof. The functional $f_A[u]$ is bounded below. Suppose a sequence $\{u_n | u_n \in K(\mathcal{G})\}$ minimizes this functional, whereby $|A^{u_n}| \leqslant |A|\ \forall n$. Then, according to (6), (7),

$$|w_n|_1 \equiv |u_n - I|_1 = |u_n|_1 \leqslant |A| + |A^{u_n}| \leqslant 2|A|.$$

Considering relations (9) and (10), we conclude from this that for all n

$$\|w_n\|^2 \leqslant 4\left(\frac{4}{(D-2)^2} + 1\right)|A|^2 \equiv b^2 < \infty\ . \tag{11}$$

The bounded sequence $\{w_n\}$ contains a weakly convergent subsequence. Suppose w_0 is a limit point of it. We choose a positive weight function p in R^D such that $x^2 p(x) \xrightarrow[|x|\to\infty]{} 0$. Let $L_2(p)$ be the space of square-integrable functions on R^D with weight p. The imbedding $L_2^1 \subset L_2(p)$ is compact, and hence the subsequence $\{w_n\}$ converges to w_0 strongly in the metric of $L_2(p)$, and hence almost everywhere in R^D. The equalities $u_0 u^+ = I$, $\det u = 1$ are also satisfied almost everywhere. This implies that $(I + w_0) \in K(\mathcal{G})$. By Fatou's lemma,

$$f_A[u_0] = \int (\partial_\mu u_0 \partial_\mu u_0^+)\, d^D x \leqslant \varliminf_{n\to\infty} f_A(u_n)\ . \tag{12}$$

Since the sequence $\{u_n\}$ is a minimizing sequence, $\varliminf f_A[u_n] = \inf_{u \in K(\mathcal{G})} f_A[u]$. Thus, u_0 provides an absolute minimum of the functional f_A, which was required to prove.

3. We now turn to the formulation of the quantum theory of a gauge field in D-dimensional Euclidean space in terms of a path integral. We restrict ourselves to integration over fields A_μ, belonging to the space L_2, although for some applications this condition may be too restrictive. By an orbit we shall understand a set of fields which are obtained from some one field by means of transformations belonging to the Sobolev gauge group $K(\mathcal{G})$. In writing the path integral, the gauge conditions must be introduced in such a manner that one and only one field is considered on each orbit. In correspondence with Proposition 2 on each orbit there is a field $A_\mu^{u_0}$ corresponding to an absolute minimum of the functional

$f_A[u] \equiv |A^u|^2$. It can be thought that in the general case on a fixed orbit this field is unique, since the Sobolev group does not contain global transformations by constant matrices. It is also reasonable to suppose that those exceptional orbits on which more than one field corresponds to an absolute minimum make a negligible contribution to the path integral. We adopt these assumptions and write the path integral in the form

$$I = \int_{\mathcal{B}} \mathcal{D} A^a_\mu \, \delta(\partial_\mu A_\mu) \, \Delta[A] \, exp(-S),$$

where

$$\Delta[A] = det\, M = det(\partial_\mu \mathcal{D}_\mu(A)),$$

and integration goes over the set $\mathcal{B}$ of all fields corresponding to an absolute minimum of the functional $f_A[u] = |A^u|^2$ on its orbits:

$$\mathcal{B} = \{ A_\mu \,|\, A_\mu \in L_2,\ |A| \leq |A^u| \ \forall\, u \in K(\mathcal{G}) \} . \tag{13}$$

It has been noted that fields $A_\mu \in \mathcal{B}$ satisfy the condition $\partial_\mu A_\mu = 0$.

In the original work [1] integration was taken over a broader set of fields C_0 (the notation of [1])

$$C_0 = \{ A_\mu \,|\, A_\mu \in L_2,\ \partial_\mu A_\mu = 0,\ \ \partial_\mu \mathcal{D}_\mu(A) \geq 0 \} . \tag{14}$$

This set contains fields corresponding not only to absolute minima of the functional $f_A[u]$, but also to relative minima of it on its orbits, as is evident from Proposition 1. It will be shown below that there are relative minima which are not absolute minima. Although the set C_0 is too broad, the inclusion $\mathcal{B} \subset C_0$ makes it possible to obtain useful estimates.

We note some properties of the sets $\mathcal{B}$ and C_0. First of all, $\mathcal{B}$ and C_0 are invariant under a transformation of fields A_μ by elements of the Euclidean D-dimensional Poincaré group $O(D) \times T^D$ including reflections. Second, $\mathcal{B}$ and C_0 are invariant under scale transformations

$$A_\mu(x) \longrightarrow A'_\mu(x) = \beta A_\mu(\beta x) , \tag{15}$$

where β is any positive number. This follows from the absence in definitions (13), (14) of dimensional constants and can be verified directly. Finally, the sets $\mathcal{B}$ and C_0 are convex. In other words, if $A_\mu \in \mathcal{B}$, $A'_\mu \in \mathcal{B}$, $\alpha > 0$, $\alpha' > 0$, $\alpha + \alpha' = 1$, then $(\alpha A_\mu + \alpha' A'_\mu) \in \mathcal{B}$ (and similarly for the set C_0). Convexity of the set $\mathcal{B}$ follows from definition (13) and the linearity of the condition

$$|A^u|^2 - |A|^2 = \int d^D x \{ (\partial_\mu u, \partial_\mu u) - 2(A_\mu, u \partial_\mu u^+) \} \geq 0$$

with respect to A_μ. Similarly, convexity of C_0 follows from the linearity of the relations $\partial_\mu A_\mu = 0$, $\partial_\mu \mathcal{D}_\mu(A) \geq 0$ in A_μ. The equation $\partial_\mu A_\mu = 0$ defines a linear subspace in which the sets C_0 and $\mathcal{B}$ lie.

The field $A_\mu' = 0$ is contained in the convex set $\mathcal{B}$. Hence, if some field A_μ belongs to the boundary of the set $\mathcal{B}$, then $\alpha A_\mu \in \mathcal{B}$ for $0 < \alpha \leq 1$, and, conversely, $\alpha A_\mu \notin \mathcal{B}$ for

$\alpha > 1$. The set C_0 has a similar property. Using this, we shall show that not every relative minimum of the functional $f_A[u]$ is an absolute minimum. Suppose A_μ belongs to the boundary $\Gamma(C_0)$ of the set C_0. Then according to Proposition 1 the first two variations of the functional $f_A[u]$ about the point $u = I$ vanish. The third variation

$$\delta^3 f_A = \frac{1}{3}\int d^D x\,(A_\mu, [\omega, [\omega, \partial_\mu \omega]]),$$

equal to the cubic term of the expansion of the function $f_A[e^{-\omega}]$ in a series in ω is, generally speaking, nonzero. Hence, in the general case the functional $f_A[u]$ has no relative minimum for $u = I$ when $A_\mu \in \Gamma(C_0)$. Moreover, there exists a matrix u_0 for which $f_A[u_0] < f_A[I]$. We go over to the field $(1 - \varepsilon)A_\mu$, where ε is positive and arbitrarily small. The field $(1 - \varepsilon)A_\mu$ lies inside C_0, and hence the functional $f_{(1-\varepsilon)A}[u]$ has a relative minimum for $u = I$. Now from considerations of continuity $f_{(1-\varepsilon)A}[u^0] < f_{(1-\varepsilon)A}[I]$. Hence, this relative minimum is not an absolute minimum, and $\mathcal{B} \neq C_0$.

In conclusion, we note that the invariance of the set $\mathcal{B}$ under the dilation (15) is closely connected with the suppression of low-frequency fields which was indicated by Gribov. If $A_\mu(x)$ lies on the boundary of the set $\mathcal{B}$, then the field $\beta A_\mu(\beta x)$ also lies on the boundary. For $\beta < 1$ the transformed field varies more slowly than the original, and the magnitude of the field $\beta A_\mu(\beta x)$ is less than the magnitude of the field A_μ. Hence, the restrictions imposed by the condition $A_\mu \in \mathcal{B}$ are strengthened with a decrease of the frequency.

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